Identifying Houston Road Areas with Recurrent Flooding and Pothole Complaints Using 311 Data

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Houston, Texas | 2025 and 2026 YTD | ZIP-code hotspot screening

Background

Houston road maintenance is often reactive, meaning roadway issues are usually addressed after residents file complaints. Repeated complaints about flooding and potholes can act as a low-cost signal for places where drainage-related problems and pavement distress may be concentrated.

Research Question

Which Houston ZIP-code areas show recurring flooding and pothole complaints, and which overlap areas should be prioritized for maintenance inspection?

Objective

Identify flooding hotspots, identify pothole hotspots, detect ZIP codes where both complaint types occur together, and rank those overlap areas using a simple priority score.

Scope

- Study area: Houston, Texas
- Study period: 2025 and 2026 YTD
- Complaint types analyzed: Flooding and Pothole
- Spatial unit: ZIP code
- Data source: City of Houston 311 public service request data

Data Source

Official City of Houston 311 public service request data

https://houstontx.gov/311/servicerequestdata.html

Variables Used

Raw variables

Case ID, Created Date Local, Closed Date, Incident Case Type, Latitude, Longitude, ZIP Code, Status, Incident Address

Derived variables

Request year, request month, valid coordinate flag, overlap flag, priority score

Methodology

- Downloaded the 2025 and 2026 YTD Houston 311 public data files.
- Combined both yearly datasets into one working table.
- Filtered the records to keep only Flooding and Pothole complaints.
- Standardized dates, ZIP codes, and coordinates.
- Removed duplicate case IDs and invalid spatial records from mapping.
- Aggregated complaint counts by ZIP code and ranked overlap areas.

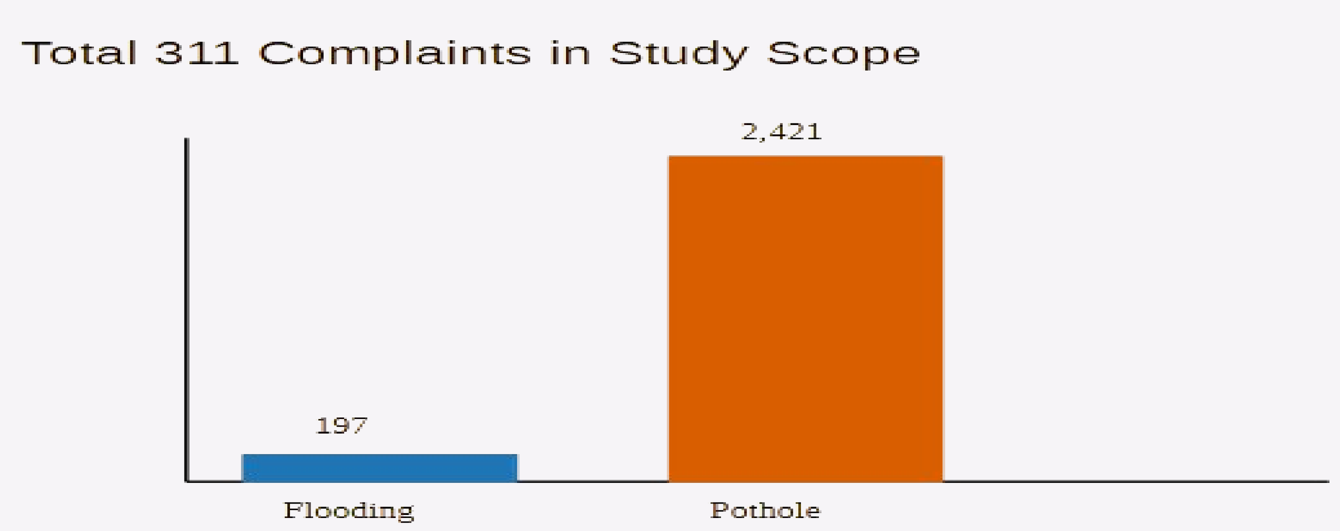
Hotspot / Overlap Criteria

Hotspot definition: a ZIP code with repeated complaints of one type.

Overlap definition: a ZIP code with at least one flooding complaint and at least one pothole complaint during the study period.

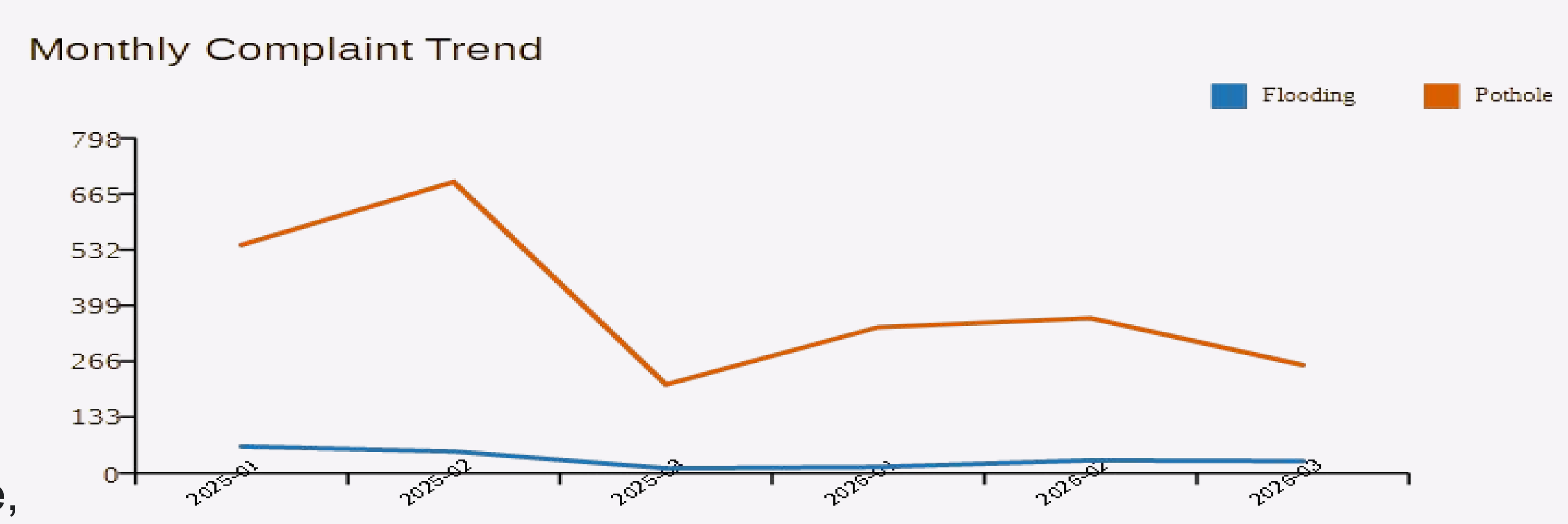
Priority score: 2 x Flooding Count + 1 x Pothole Count + 10 x Overlap Flag

Visual 1: Complaint Totals



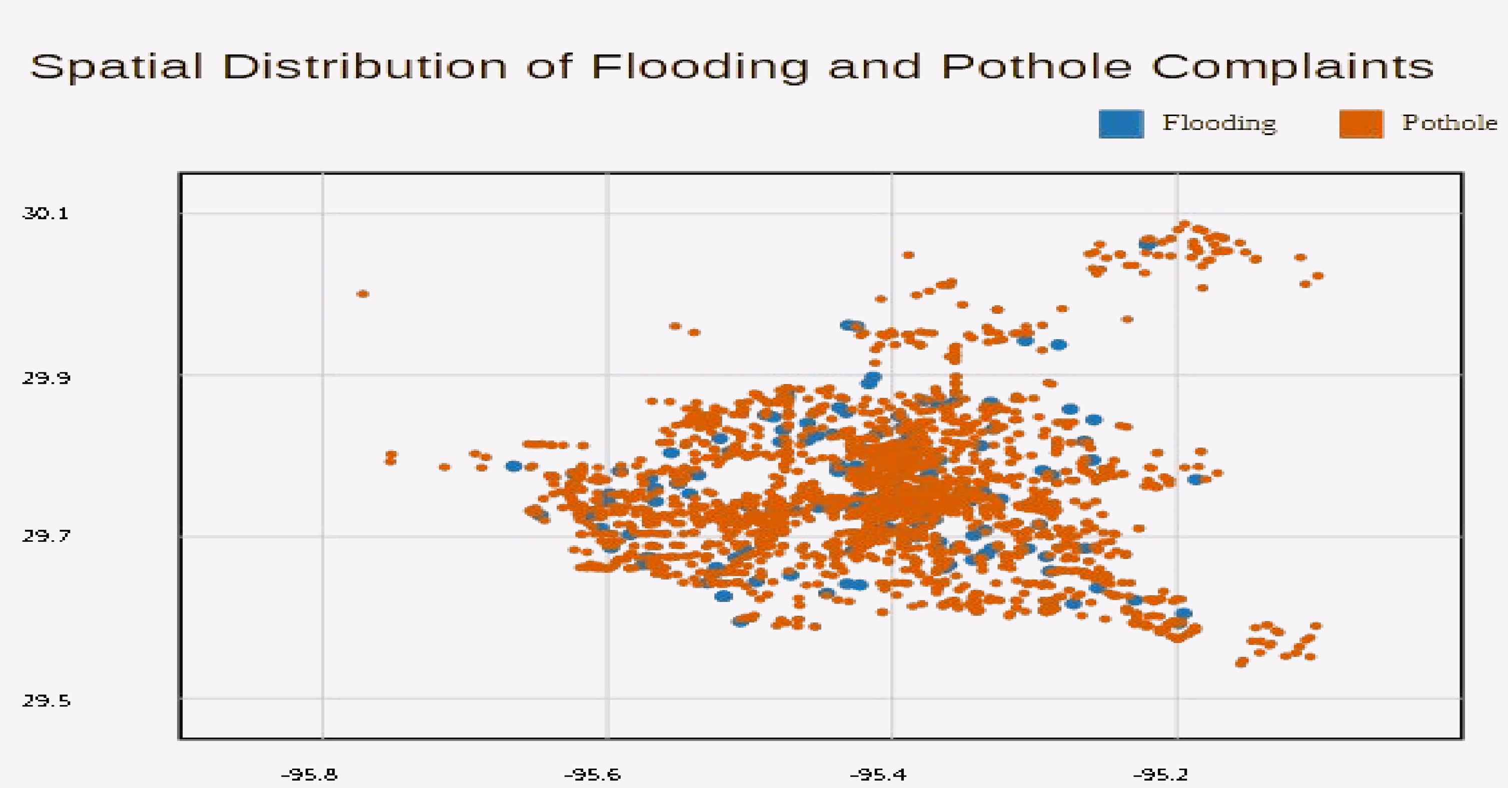
This chart compares the total number of flooding and pothole complaints in the filtered study dataset.

Visual 2: Monthly Trend



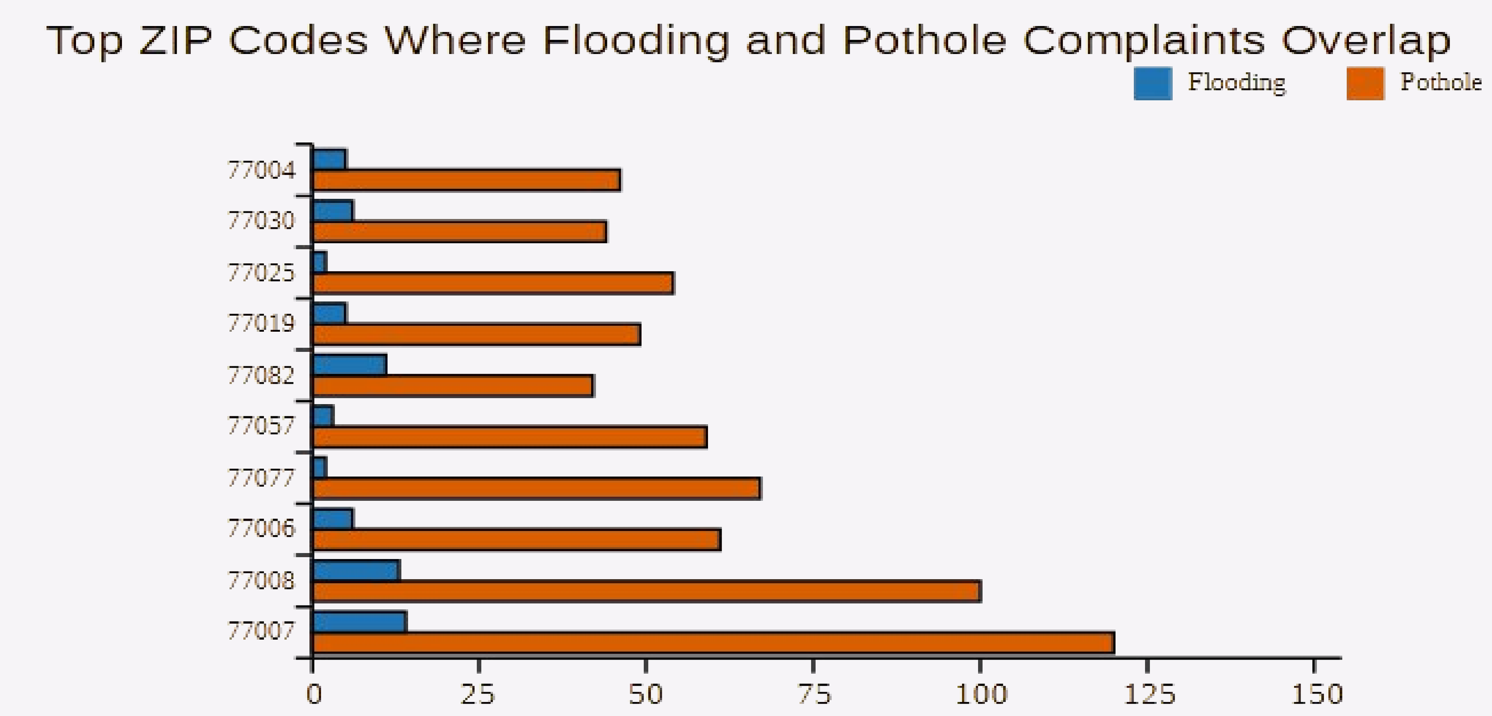
Both complaint types peaked in February, showing a concentrated reporting period within the selected study window.

Visual 3: Spatial Distribution



The mapped complaint locations are not evenly distributed across Houston and instead form visible spatial clusters.

Visual 4: Overlap ZIP Codes



ZIP codes 77007 and 77008 showed the strongest overlap between flooding and pothole complaints.

Results

- Raw records loaded: 173,977
- Filtered study records: 2,618
- Flooding complaints: 197
- Pothole complaints: 2,421
- Valid mapped records: 2,617
- ZIP codes with both complaint types: 62
- Peak flooding month: February (81)
- Peak pothole month: February (1,063)

Insights

- Pothole complaints were much more frequent than flooding complaints during the study period.
- Complaint overlap provides a stronger screening signal than isolated single-issue complaints.
- ZIP codes 77007, 77008, 77006, 77077, and 77057 had the highest overlap priority scores.
- Overlap areas may indicate places where pavement distress and drainage-related issues coexist.

Conclusion

Houston 311 complaint data can be used as a low-cost screening tool to identify flooding and pothole hotspots and to highlight overlap ZIP codes for priority roadway and drainage inspection.

Limitations

- 311 data reflects reported complaints rather than all real-world infrastructure defects.
- ZIP code is a simplified spatial unit and does not capture exact road segments.
- Overlap between flooding and pothole complaints indicates association, not causation.
- Field inspection would still be required before design or maintenance decisions are made.